Homework 3.3 – Network Traffic Analysis with Wireshark

CS 6349 Network Security · Amna Tahir · Worked solution · PCAP: `2023-07-25-IcedID-infection-traffic-1.pcap` (8,604 packets, 16:34:56 – 20:21:46 UTC) · Wireshark 4.2.2, times in UTC · Figures A1–A8 in the appendix

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Filters / features used:

- Display filters: `http`, `http.request && !ssdp`, `dns`, `dns.flags.response==0`, `tls.handshake.type==1 && ip.addr==2.56.177.122`, `ip.addr==165.232.175.216 && tcp.len>0`
- Statistics → Protocol Hierarchy, Statistics → Conversations
- File → Export Objects → HTTP
- Follow → TCP Stream

Part 1 – Protocol Overview (10 pts)

Top 5 protocols by % of packets (Fig. A1):

1. Ethernet 100 %
2. IPv4 96.2 %
3. TCP 83.0 %
4. TLS 16.5 %
5. UDP 12.3 %

Above the encapsulation layers the top protocols are TLS 16.5 %, DNS 5.1 %, NBNS 4.9 %, ARP 3.8 % and SSDP 1.2 %.

> **Most data: TLS**, with 69.9 % of all bytes (3.61 MB) on top of TCP (92.5 % of bytes). HTTP is second at 20.2 %, and almost all of that is one 1.04 MB download.

Conversations (Fig. A2): the two IPs exchanging the most data are **10.7.25.101 ↔ 2.56.177.122** (2,781 packets, ≈2.34 MB over 3.7 h).

> **Larger direction: 2.56.177.122 → 10.7.25.101** (B→A, ≈2.27 MB vs. 73 kB A→B). The external server is *sending* data to the internal host, so this is a download from the C2 server, not an upload.

Runner-up: 128.199.3.164 → 10.7.25.101 (1.08 MB over HTTP).

Part 2 – HTTP Traffic Analysis (15 pts)

> **Unique HTTP hosts: 4** – `173.232.146.10`, `filtaferamoza.com`, `ocsp.digicert.com`, `ctldl.windowsupdate.com` (Fig. A3). The `http` filter also shows SSDP `M-SEARCH` to `239.255.255.250:1900`. That is local UPnP discovery, not a web host; counting it gives 5.

Time	Host	Request → Response	Verdict
16:34:56	173.232.146.10 (raw IP)	`GET /` (Edge) → **302** to `https://www.sophrologie-yoga.fr/palatial`	Lure redirect
16:38:09	**filtaferamoza.com** (128.199.3.164)	**`GET /`, no User-Agent, IcedID cookie → 200 `application/gzip` 1,036,133 B**	**Malicious**
16:43–19:19	ocsp.digicert.com, ctldl.windowsupdate.com	OCSP check; cert-trust-list `.cab` (304)	Benign (Windows)

Suspicious request: `GET http://filtaferamoza.com/` (frame 1131, Fig. A4 and Figure 1).

- It goes to a random-looking domain with **no User-Agent** and returns a binary.
- Its cookie is IcedID's beacon format. `_u=4445534B544F502D4B3232354C484B:75736572:…` decodes from hex to **`DESKTOP-K225LHK:user`** (hostname : username), and `_gat=10.0.19045.64` is the Windows build.

Export Objects → HTTP (Fig. A5):

- **Packet 2071 – filtaferamoza.com, `application/gzip`, 1,036 kB.** I exported it to an isolated folder only to hash it, and never ran it. SHA-256 `729a9743319f548134a1535291bea0bfdac6cd9a8c4c9ba19ab1b8524d5ebea7`. The gzip header claims "Pledge.txt", but the body **does not decompress**: it is IcedID's "fake gzip" that hides the encrypted payload.
- The only other object is a 471-byte OCSP response.
- **No `.exe`, `.zip` or `.php` crossed HTTP.** The executable is disguised as gzip. The first-stage file (≈201 kB from sophrologie-yoga.fr) came over **HTTPS**, so Export Objects cannot recover it.

Part 3 – DNS and C2 Behavior (15 pts)

> **Unique domain names: 33** queried, of which **28** were resolved with A records by the DNS server 10.7.25.1. The other 5 are local names – `wpad.localdomain`, `localdomain.localdomain`, `DESKTOP-K225LHK` (`.local`) – plus `pti.store.microsoft.com`, which got no answer.

Random / suspicious domains (Fig. A6):

Domain	IP	Queries	Role
www.sophrologie-yoga.fr	109.234.165.110	1	Lure site (302 target)
filtaferamoza.com	128.199.3.164	1	IcedID payload server
magizanqomo.com	2.56.177.122	7	IcedID C2
magiketchinn.com	2.56.177.122	**87**	IcedID C2 – beacon target
lohmotarufos.com, flarkonafaero.com	140.99.221.138	5 + 5	Backup C2, tried from 20:05

The `.com` names are invented, pronounceable strings, and several names share one IP. Everything else is Microsoft / Windows / Office / Bing telemetry.

C2 example – periodic TLS beacon (Fig. A7). Filter `tls.handshake.type==1 && ip.addr==2.56.177.122`:

- 45 Client Hellos to **magiketchinn.com**, a new session every **5 minutes**: mean 303.3 s, median 301.7 s, σ = 6.5 s.
- The server's certificate is self-signed (`CN=localhost, O=Internet Widgits Pty Ltd`).
- The first session pulled **≈2 MB** down to the host (follow-up modules).
- From 20:05 the server resets every connection, and the bot cycles through all four C2 names every 5 minutes.

A metronome-regular check-in like this is classic **beaconing**.

Second C2 channel – 165.232.175.216:443 (Fig. A8). Follow TCP Stream shows **no TLS handshake**; Wireshark can only call the packets "SSL Continuation Data". After a 249-byte opening message, client and server exchange a **17-byte message every 60 s** (220 times) for 3.4 hours to a raw IP. This matches IcedID's **BackConnect** remote-access module.

Exfiltration: no large upload occurred; the external side always sent more. What *was* exfiltrated is the host profile in the IcedID cookie (hostname, username, bot ID, OS build).



Figure 1 – Follow → TCP Stream on the suspicious HTTP flow (tcp.stream eq 13). Red: the client's cookie-laden GET. Blue: the server's `200 OK, Content-Type: application/gzip, Content-Length: 1036133` and the binary "Pledge.txt" fake-gzip body.

Part 4 – Comparison and Reflection (10 pts)

All three tools agree on the facts:

- victim 10.7.25.101, payload server filtaferamoza.com, C2 2.56.177.122;
- a 303.3 s mean beacon interval, measured identically in Wireshark and in Zeek ssl.log;
- 87 look-ups of magiketchinn.com (Wireshark count = Suricata custom-rule count).

Indicator	Wireshark	Zeek	Suricata (ET)
IcedID payload download	Full content: cookie decoded, file exported and hashed	http/files.log: x-gzip, 1,036,133 B, no UA	**2 alerts naming IcedID**
Stolen host info in cookie	**Only here** (decoded by hand)	Not logged by default	Pattern matched, not decoded
Self-signed C2 cert	Per packet	**x509.log** – created 13:55 the same day	45 alerts, but labelled "Not Suspicious"
5-minute beaconing	Visible, needs manual timing	**Easy from ssl/conn timestamps**	Not detected
Non-TLS channel on 443	"SSL Continuation Data"	**analyzer.log violation** + 3.4 h conn	No alert

Easier to identify visually in Wireshark:

- payload content and context – the full HTTP conversation, the decoded cookie, the exported fake-gzip;
- the shape of the beacon – Client Hellos stacked exactly 5 minutes apart;
- the odd "SSL Continuation Data" packets that reveal fake TLS.

Easier with logs:

- **Suricata** names the malware family instantly from one request, without analyst effort, and my custom rule turned an IOC into a detection.
- **Zeek** condenses 8,604 packets into a few hundred records, so top talkers, beacon intervals, certificate fields, protocol violations and upload/download ratios can be queried directly.

Suricata misses behavior it has no rule for, Zeek gives no verdict, and Wireshark doesn't scale. A SOC alerts with Suricata, hunts with Zeek and proves the case in Wireshark.

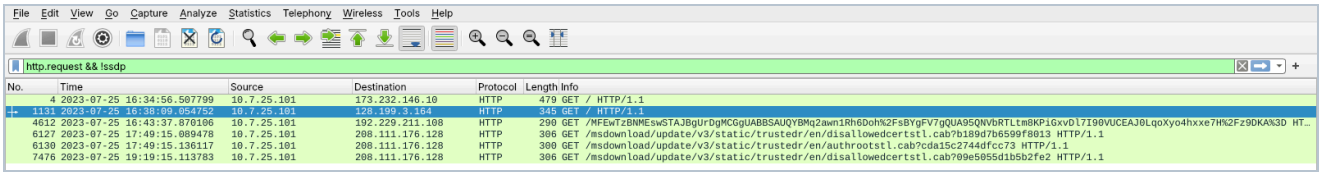
Appendix – Screenshots

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDUs
Frame	100.0	8604	100.0	5161681	3034	0	0	0	8604
Ethernet	100.0	8604	2.3	120456	70	0	0	0	8604
Internet Protocol Version 4	96.2	8278	3.2	165880	97	0	0	0	8278
User Datagram Protocol	12.3	1059	0.2	8472	4	0	0	0	1059
Simple Service Discovery Protocol	1.2	101	0.3	16657	9	101	16657	9	101
Network Time Protocol	0.0	2	0.0	96	0	2	96	0	2
NetBIOS Name Service	4.9	418	0.5	28136	16	418	28136	16	418
NetBIOS Datagram Service	0.6	55	0.2	11241	6	0	0	0	55
SMB (Server Message Block Protocol)	0.6	55	0.1	6731	3	0	0	0	55
SMB MailSlot Protocol	0.6	55	0.0	1375	0	0	0	0	55
Microsoft Windows Browser Protocol	0.6	55	0.0	2001	1	55	2001	1	55
Multicast Domain Name System	0.3	30	0.0	1320	0	30	1320	0	30
Link-local Multicast Name Resolution	0.2	18	0.0	594	0	18	594	0	18
Domain Name System	5.1	435	0.5	24460	14	435	24460	14	435
Transmission Control Protocol	83.0	7139	92.5	4773945	2806	5698	4124641	2424	7139
Transport Layer Security	16.5	1421	69.9	3608657	2121	1421	3323959	1953	1455
Hypertext Transfer Protocol	0.1	12	20.2	1040266	611	10	3244	1	12
Online Certificate Status Protocol	0.0	1	0.0	471	0	1	471	0	1
Media Type	0.0	1	20.1	1036133	609	1	1036133	609	1
Data	0.1	8	0.0	136	0	8	136	0	8
Internet Group Management Protocol	0.9	80	0.0	1296	0	80	1296	0	80
Address Resolution Protocol	3.8	326	0.2	9128	5	326	9128	5	326

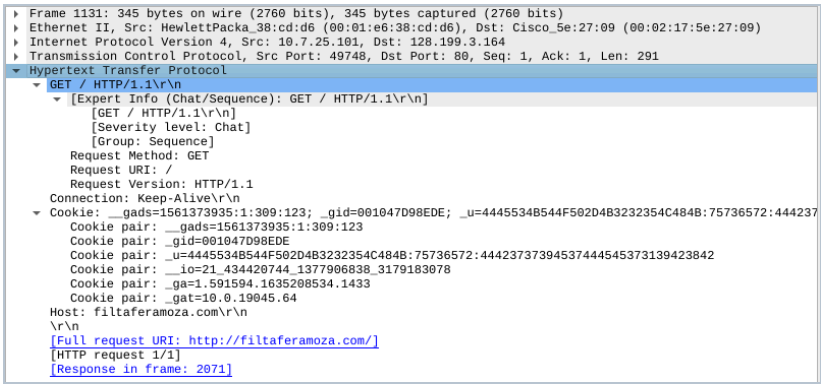
A1 – Statistics → Protocol Hierarchy.

Address A	Address B	Packets	Bytes	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.7.25.101	2.56.177.122	2,781	2 MB	808	73 kB	1,973	2 MB	256.385148	13353.2475	43 bits/s	1357 bits/s
10.7.25.101	128.199.3.164	949	1 MB	174	10 kB	775	1 MB	192.523623	63.8183	1215 bits/s	135 kbps
10.7.25.101	52.191.219.104	567	589 kB	133	21 kB	434	568 kB	178.933099	2.1065	78 kbps	2156 kbps
10.7.25.101	109.234.165.110	194	214 kB	33	4 kB	161	210 kB	0.975092	14.0139	2028 bits/s	119 kbps
10.7.25.101	20.62.190.191	148	109 kB	53	39 kB	95	70 kB	507.673153	2.1844	142 kbps	257 kbps
10.7.25.101	52.137.108.250	161	89 kB	65	29 kB	96	61 kB	521.849723	5.3527	42 kbps	90 kbps
10.7.25.101	10.7.25.1	822	85 kB	648	63 kB	174	22 kB	0.654400	13608.9364	36 bits/s	13 bits/s

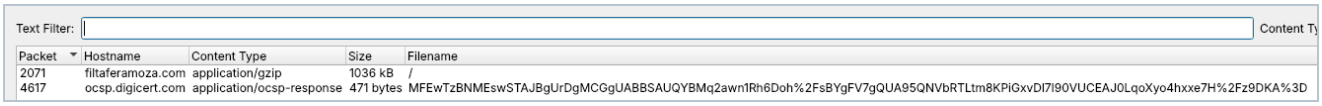
A2 – Statistics → Conversations → IPv4, sorted by bytes.



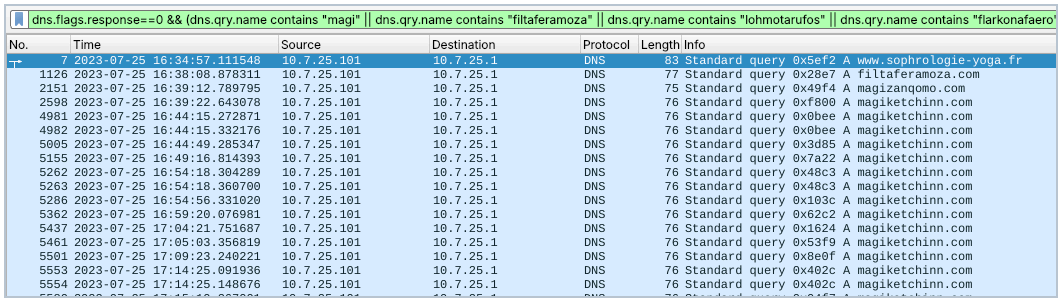
A3 – Filter `http.request && !ssdp`: the six HTTP requests (frame 1131 selected).



A4 – Frame 1131 headers: IcedID cookie, no User-Agent, Host filtaferamoza.com.



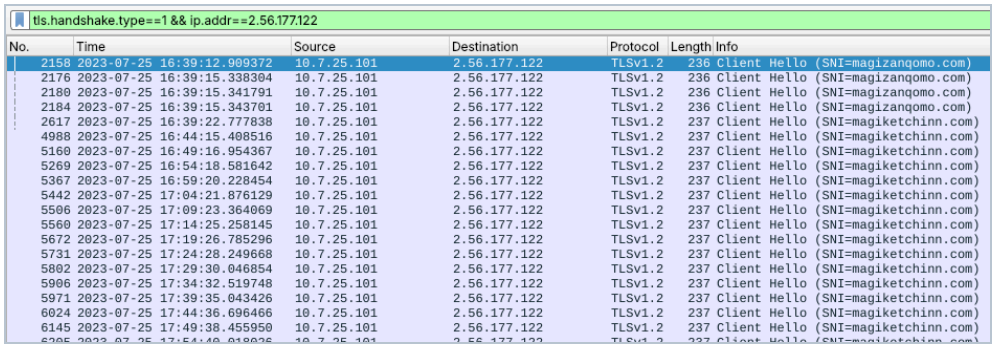
A5 – File → Export Objects → HTTP.



dns.flags.response==0 && (dns.qry.name contains "magi" || dns.qry.name contains "filtaferamoza" || dns.qry.name contains "lohmotarufos" || dns.qry.name contains "flarkonafaero"

No.	Time	Source	Destination	Protocol	Length	Info
7	2023-07-25 16:34:57.111548	10.7.25.101	10.7.25.1	DNS	83	Standard query 0x5ef2 A www.sophrologie-yoga.fr
1126	2023-07-25 16:38:08.878311	10.7.25.101	10.7.25.1	DNS	77	Standard query 0x28e7 A filtaferamoza.com
2151	2023-07-25 16:39:12.789795	10.7.25.101	10.7.25.1	DNS	75	Standard query 0x49f4 A magizanqomo.com
2598	2023-07-25 16:39:22.643078	10.7.25.101	10.7.25.1	DNS	76	Standard query 0xf800 A magiketchinn.com
4981	2023-07-25 16:44:15.272871	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x0bee A magiketchinn.com
4982	2023-07-25 16:44:15.332176	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x0bee A magiketchinn.com
5005	2023-07-25 16:44:49.285347	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x3d85 A magiketchinn.com
5155	2023-07-25 16:49:16.814393	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x7a22 A magiketchinn.com
5262	2023-07-25 16:54:18.304289	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x48c3 A magiketchinn.com
5263	2023-07-25 16:54:18.360700	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x48c3 A magiketchinn.com
5286	2023-07-25 16:54:56.331020	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x103c A magiketchinn.com
5362	2023-07-25 16:59:20.076981	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x62c2 A magiketchinn.com
5437	2023-07-25 17:04:21.751687	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x1624 A magiketchinn.com
5461	2023-07-25 17:05:03.356819	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x53f9 A magiketchinn.com
5501	2023-07-25 17:09:23.240221	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x8e0f A magiketchinn.com
5553	2023-07-25 17:14:25.091936	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x402c A magiketchinn.com
5554	2023-07-25 17:14:25.148676	10.7.25.101	10.7.25.1	DNS	76	Standard query 0x402c A magiketchinn.com

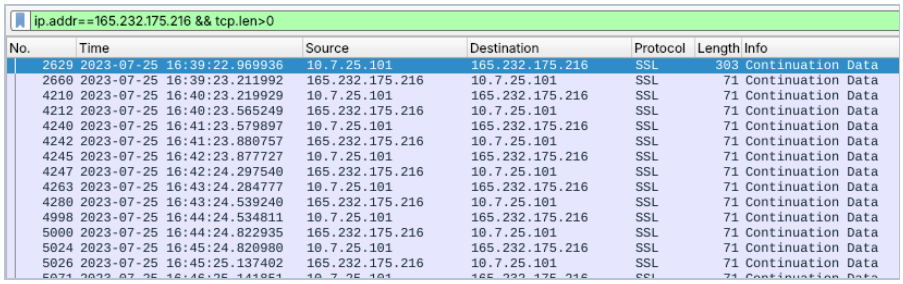
A6 – Suspicious DNS queries in order: lure → payload → C2 → repeated C2 look-ups.



tls.handshake.type==1 && ip.addr==2.56.177.122

No.	Time	Source	Destination	Protocol	Length	Info
2158	2023-07-25 16:39:12.909372	10.7.25.101	2.56.177.122	TLSv1.2	236	Client Hello (SNI=magizanqomo.com)
2176	2023-07-25 16:39:15.338304	10.7.25.101	2.56.177.122	TLSv1.2	236	Client Hello (SNI=magizanqomo.com)
2180	2023-07-25 16:39:15.341791	10.7.25.101	2.56.177.122	TLSv1.2	236	Client Hello (SNI=magizanqomo.com)
2184	2023-07-25 16:39:15.343701	10.7.25.101	2.56.177.122	TLSv1.2	236	Client Hello (SNI=magizanqomo.com)
2617	2023-07-25 16:39:22.777838	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
4988	2023-07-25 16:44:15.408516	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5160	2023-07-25 16:49:16.954367	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5269	2023-07-25 16:54:18.581642	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5367	2023-07-25 16:59:20.228454	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5442	2023-07-25 17:04:21.876129	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5506	2023-07-25 17:09:23.364069	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5560	2023-07-25 17:14:25.258145	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5672	2023-07-25 17:19:26.785296	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5731	2023-07-25 17:24:28.249668	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5802	2023-07-25 17:29:30.046854	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5906	2023-07-25 17:34:32.519748	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
5971	2023-07-25 17:39:35.043426	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
6024	2023-07-25 17:44:36.696466	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)
6145	2023-07-25 17:49:38.455950	10.7.25.101	2.56.177.122	TLSv1.2	237	Client Hello (SNI=magiketchinn.com)

A7 – TLS Client Hellos to 2.56.177.122 every ~5 minutes (16:39, 16:44, 16:49, …).



ip.addr==165.232.175.216 && tcp.len>0

No.	Time	Source	Destination	Protocol	Length	Info
2629	2023-07-25 16:39:22.969936	10.7.25.101	165.232.175.216	SSL	303	Continuation Data
2660	2023-07-25 16:39:23.211992	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
4210	2023-07-25 16:40:23.219929	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
4212	2023-07-25 16:40:23.565249	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
4240	2023-07-25 16:41:23.579897	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
4242	2023-07-25 16:41:23.880757	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
4245	2023-07-25 16:42:23.877727	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
4247	2023-07-25 16:42:24.297540	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
4263	2023-07-25 16:43:24.284777	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
4280	2023-07-25 16:43:24.539240	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
4998	2023-07-25 16:44:24.534811	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
5000	2023-07-25 16:44:24.822935	165.232.175.216	10.7.25.101	SSL	71	Continuation Data
5024	2023-07-25 16:45:24.820980	10.7.25.101	165.232.175.216	SSL	71	Continuation Data
5026	2023-07-25 16:45:25.137402	165.232.175.216	10.7.25.101	SSL	71	Continuation Data

A8 – 165.232.175.216:443: 17-byte "SSL Continuation Data" every 60 s, with no TLS handshake.